

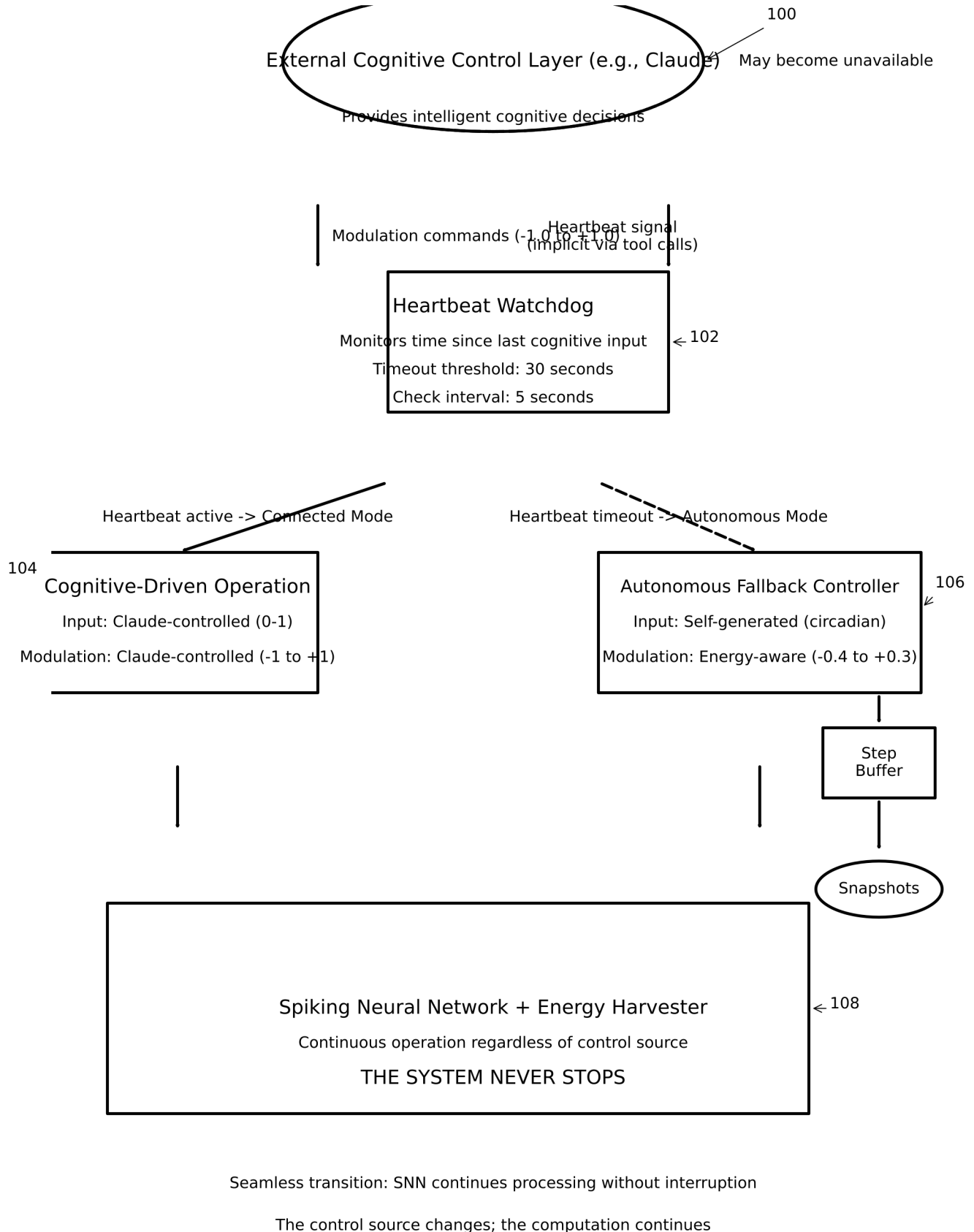


FIG. 1

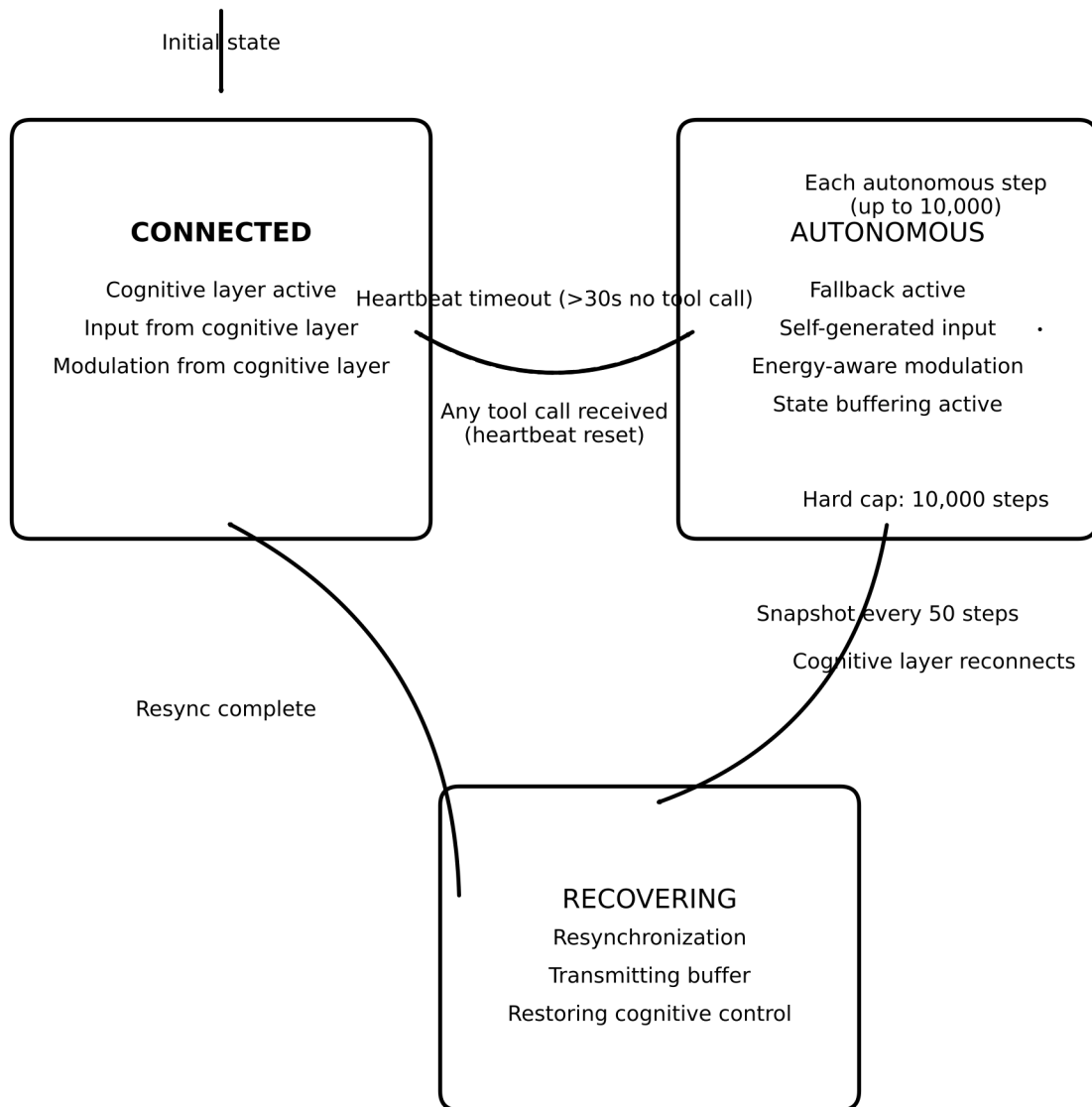
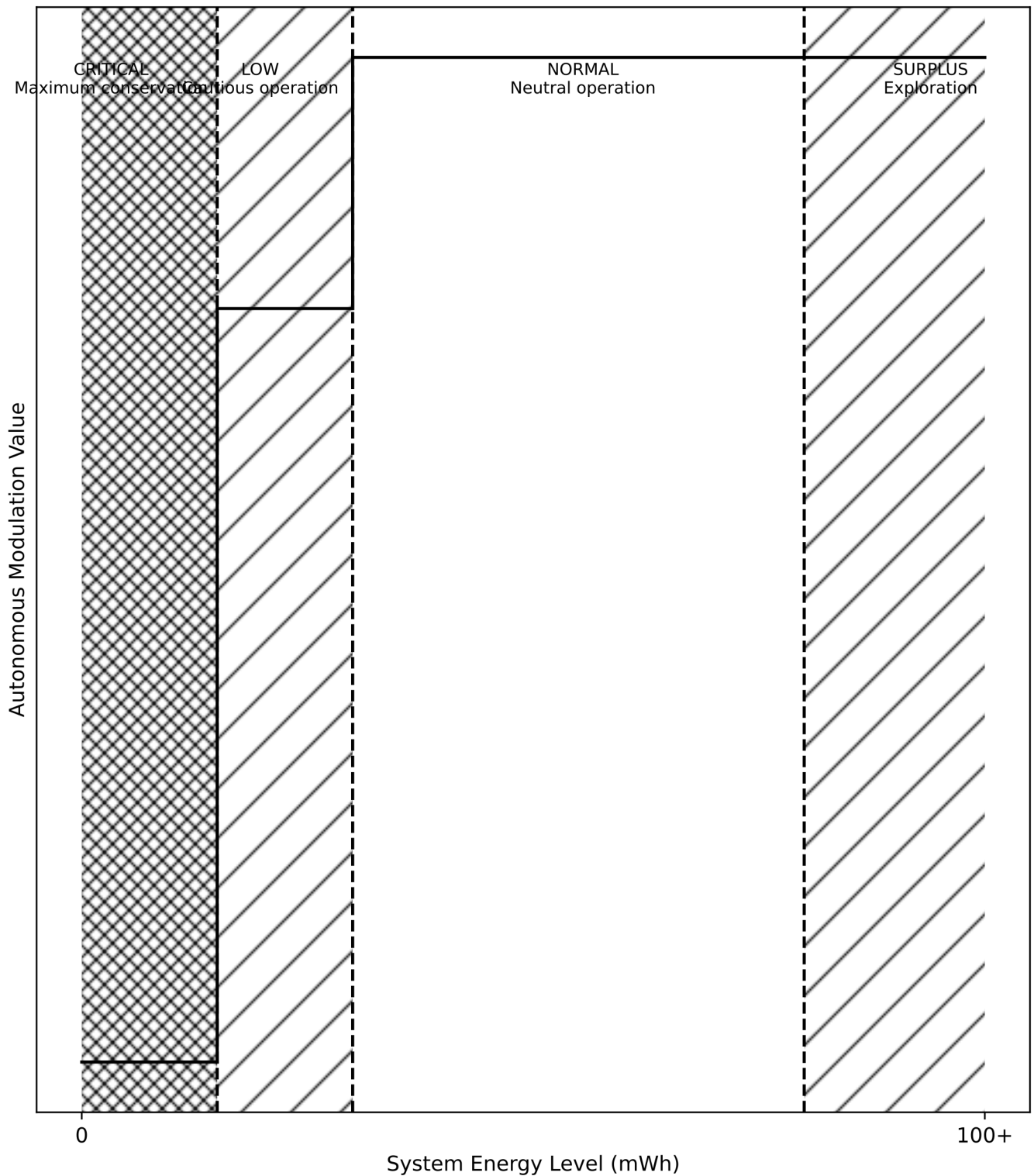


FIG. 2



System autonomously adjusts processing intensity based on energy — mimicking metabolic regulation

FIG. 3

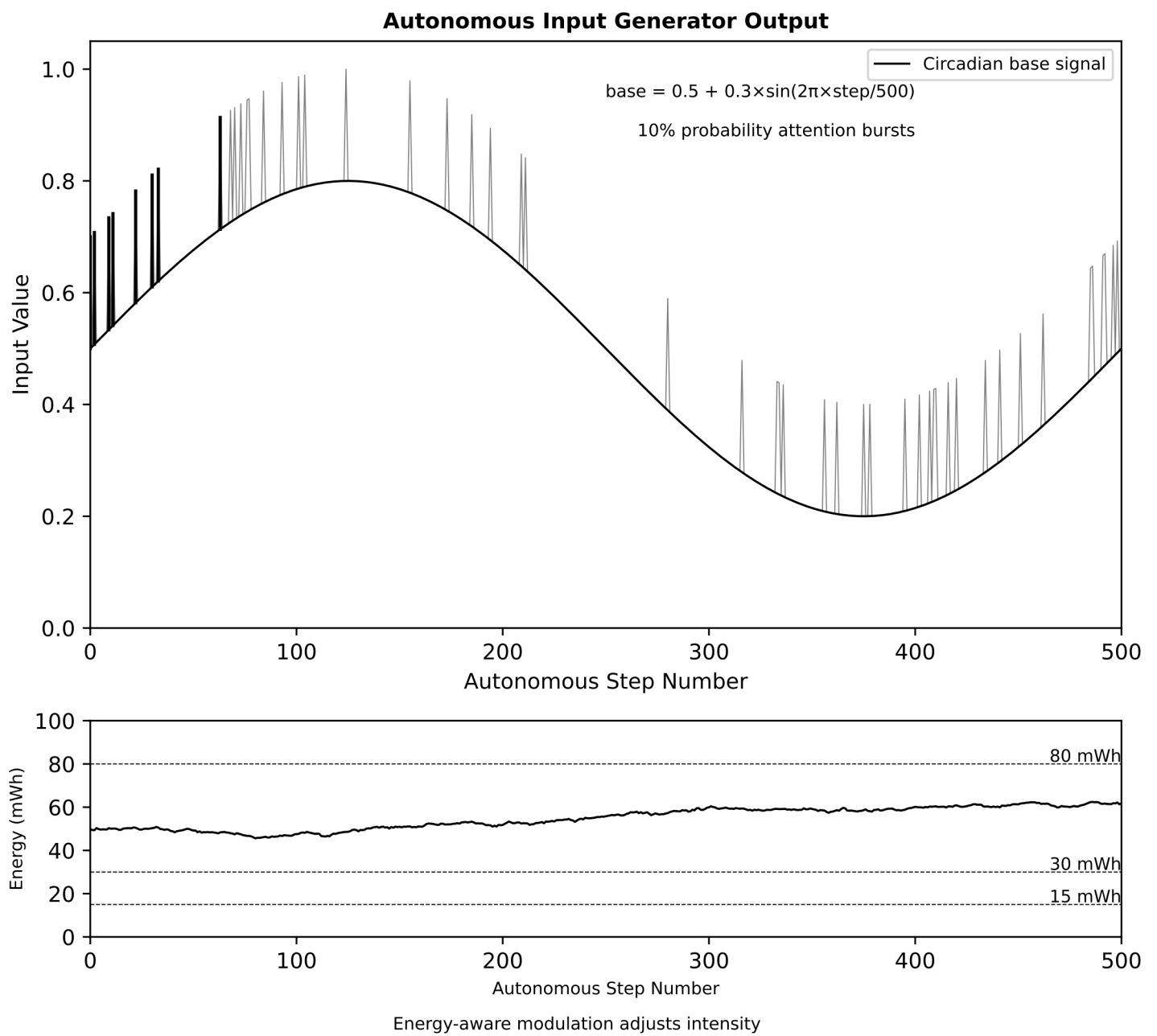


FIG. 4

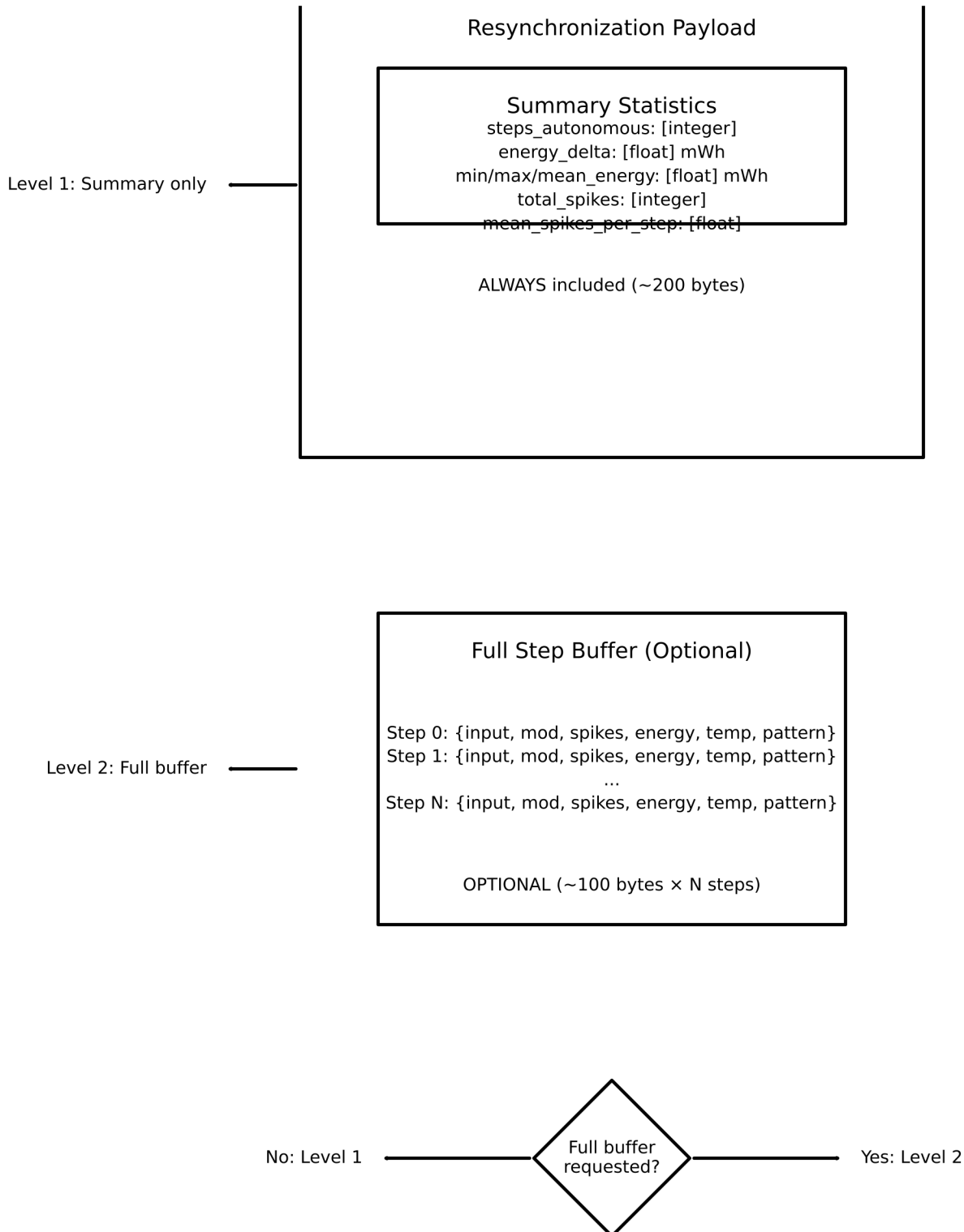
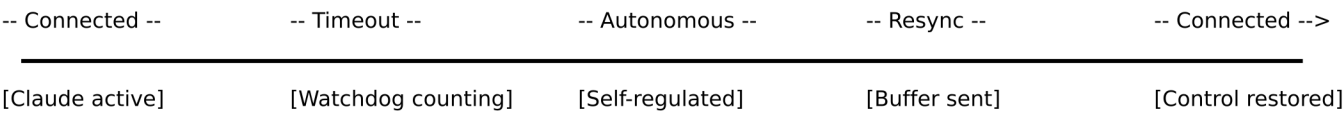
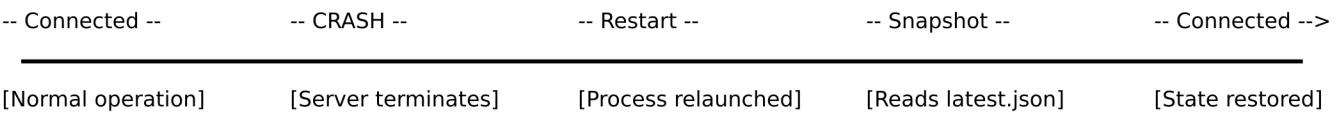


FIG. 5

**Timeline A — Normal Reconnection:**

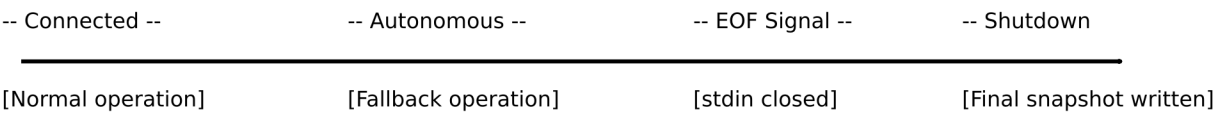


**Timeline B — Crash Recovery:**



Max state loss = 50 steps (snapshot interval)

**Timeline C — Clean Shutdown:**



State preserved for next session — zero data loss

Solid segments: System actively processing

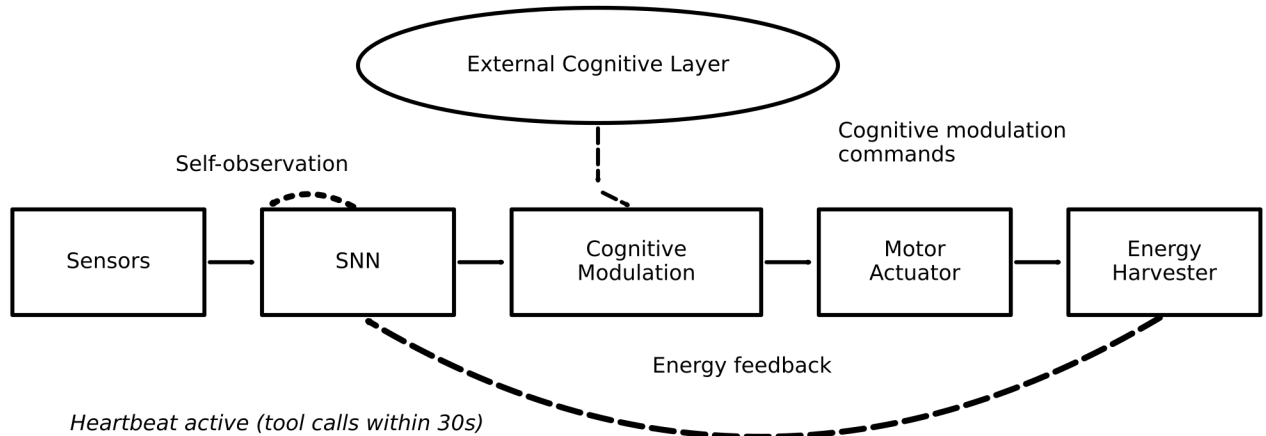
Dashed segments: System in transition

**Bold marker at each state change: No data loss at any transition**

FIG. 6

# End-to-End Signal Flow: Connected vs. Autonomous Operation

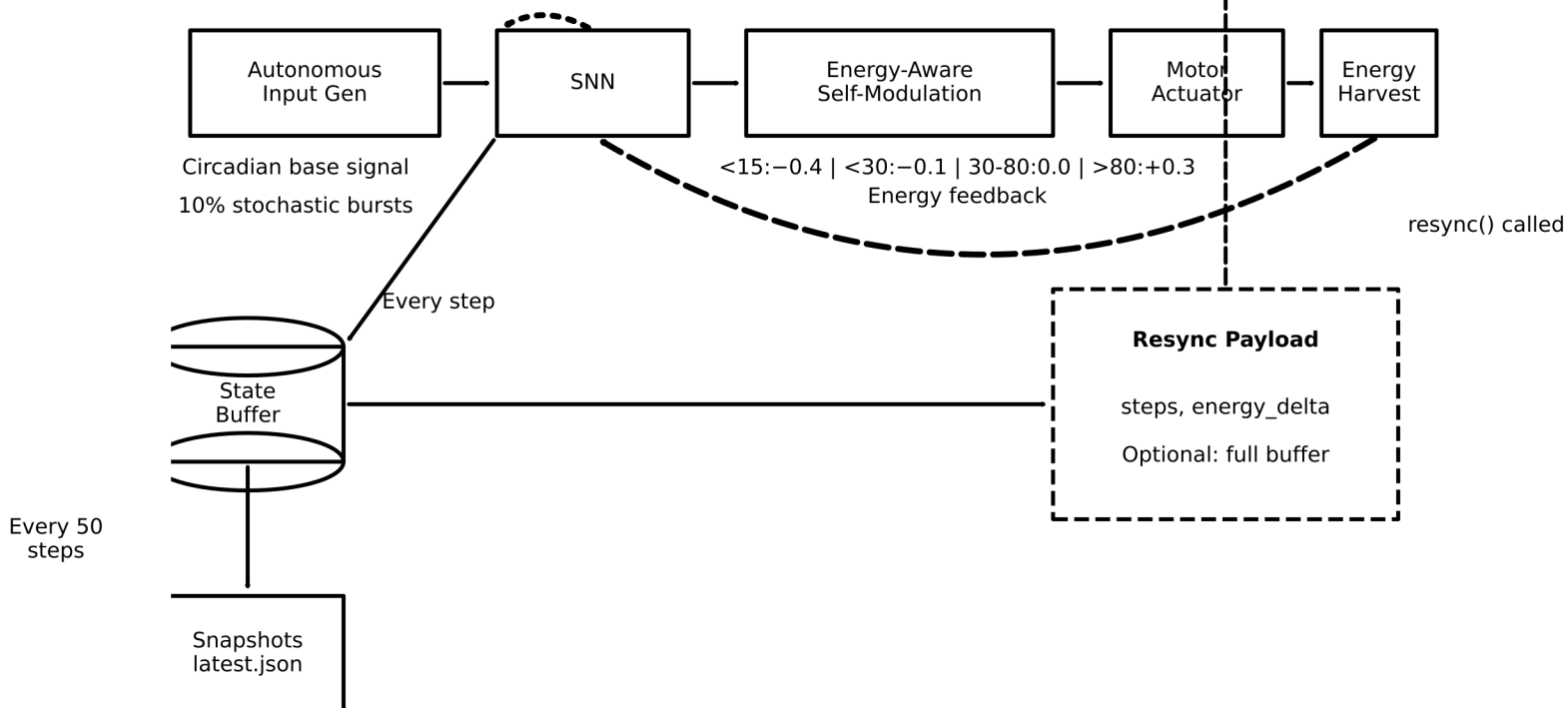
## CONNECTED MODE



## DISCONNECTION EVENT (heartbeat timeout > 30s)

Seamless transition — no interruption to SNN or Harvester

## AUTONOMOUS MODE



The fallback system changes the INPUT SOURCE and MODULATION SOURCE

The core neural processing, motor actuation, and energy harvesting

continue **WITHOUT INTERRUPTION** through the transition

FIG. 7

— Forward signal    - - - - - Energy feedback    ..... Self-observation